CONTROLS + MONITORING BLOCK DIAGRAM

ENERGY

INV-1 Hybrid Inverter
Modbus RS485: PV, grid, load, surplus

BAT-1 LFP 30 kWh
SoC, reserve 30%

Per-Circuit CT Monitor
32 ch at MDB + SDB-W

Generator
Auto-start, run hours

WATER

Tank Levels
T-1, T-2, T-3, RT-1, TT-1

Well Pump Controller
Flow, dry-run, run hours

Treatment
Filter ΔP, backwash, UV lamp status

Booster BP-1
Pressure, faults

GARDEN

Irrigation Controller
6 drip zones, IP-1 pump

Soil Moisture x 6
One per zone

Weather Station
Rain, ET, wind, irradiance

Cistern Top-Up Valve
Well via air gap

Home Controller (Home Assistant) + Zigbee/Thread + UPS
Local-first in AV rack 40S; VPN for remote; nightly backup

Phone Dashboard + Genkan Wall Tablet
Energy · Water · Pool · Climate · Alerts

PHONE DASHBOARD CONCEPT

POOL + WELLNESS

Pool Pump VS + Salt Cell
Solar-surplus schedule

Sauna + Steam
Pre-heat from app, interlocks

Cold Plunge Chiller
Off-peak / surplus

CLIMATE

VRF Gateway CU-1/2/3
Setpoints, schedules, occupancy

ERV-1, ERV-2
Boost on humidity/CO2

Dehumidifier DH-1
Gym + wet room

SAFETY + LEAK

Leak Sensors x 14
Sinks, WCs, heater, washer, AHU

Auto Shut-Off Valve
Main at 20E + flow meter

Smoke / CO + Cameras
NVR in rack

KEY AUTOMATIONS

- Pool pump + salt cell run at 10:00-15:00 when PV surplus > 2 kW; minimum 6 h turnover guaranteed
- Battery reserve 30% for outages; generator auto-start below 15% or on long outage
- Irrigation: skip if rain > 3 mm forecast/24 h or soil moisture above threshold; water at 05:00 from cisterns, well top-up only if T-1,3 < 10%
- Leak detected: close SOV, stop BP-1, push alert; daily micro-leak check by flow meter at 03:00
- Filter ΔP high: schedule backwash on PV surplus; UV lamp fault: close SOV to house + alert
- AC: zone schedules by occupancy, pre-cool on solar surplus, setback 27 °C when away; sauna/steam pre-heat from the app
- Energy: per-circuit dashboard, monthly report, alert on abnormal baseload

CONTROLS BUDGET (ROUGH THB)

TAG	ITEM	THB
HUB	Home Controller (Home Assistant on Mini-PC) + Zigbee/Thread + UPS	30,000
CTM	Per-Circuit Energy Monitor (CT Clamps, 32 ch)	28,000
INVG	Inverter + Battery Gateway (Modbus RS485)	6,000
AVG	AC/VRF Gateway (BACnet/Modbus)	25,000
POOL	Pool Pump + Salt Cell Controller	20,000
IRR	Smart Irrigation Controller (6 Zones)	12,000
WX	Weather Station (Rain, Wind)	12,000
LVL1	Level Sensors (T-1, T-2, T-3, RT-1, TT-1)	25,000
SM	Soil Moisture Sensors (6)	12,000
WP	Well Pump Controller + Flow Meter	15,000
DP	Filter ΔP Sensors + UV Status Relay	14,000
SOV	Motorized Auto Shut-Off Valve + Flow Meter	18,000
LK	Leak Sensors (14)	17,000
DASH	Wall Dashboard Tablet (Genkan) + Phone App	15,000
PRG	Wiring, Integration + Programming	90,000
	Total	339,000

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
NTS

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

CONTROLS NOTES

Local-first: automations keep running without internet.

Open protocols (Modbus, Zigbee, Matter) so devices can be swapped.

Device locations shown on the Controls layer in the 3D viewer.

SHEET TITLE

Controls + Monitoring

SHEET

E-103